

Stat 3113 Exam 3 - Practice Problems Set 1 + Key

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1. For the independent-measures t statistic, $df = (n_1 + 1) + (n_2 + 1)$.
 - a. True
 - b. False
2. An independent-measures experiment uses one sample with $n = 6$ and a second sample with $n = 10$ to compare two experimental treatments. The t statistic from this experiment will have degrees of freedom equal to the following.
 - a. 7
 - b. 8
 - c. 14
 - d. 15
3. The results of an independent-measures study produce a t statistic with $df = 36$. How many individuals participated in the entire study?
 - a. 37
 - b. 73
 - c. 38
 - d. 34
4. If both samples have $n = 10$, the independent-measures t statistic will have $df = 19$.
 - a. True
 - b. False
5. One sample with $n = 8$ and a second sample with $n = 10$ will produce an independent-measures t statistic with $df = 9$.
 - a. True
 - b. False
6. The larger the difference between the two sample means, the farther the independent-measures t statistic is from zero.
 - a. True
 - b. False
7. If the two samples are the same size, the independent measures t statistic will have an even number for degrees of freedom.
 - a. True
 - b. False
8. What of the following is true for the repeated-measures t statistic?
 - a. $df = n_1 + n_2 - 2$
 - b. $df = (n_1 - 1) + (n_2 - 1)$
 - c. $df = n - 1$
 - d. $df = n_1 + n_2 - 1$
9. A researcher uses a repeated-measures study to compare two treatment conditions with a set of 20 scores in each treatment. What would be the value of df for the repeated-measures t statistic?
 - a. $df=18$
 - b. $df=38$
 - c. $df=19$
 - d. $df=39$

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10. For a repeated-measures experiment comparing two treatment conditions, the t statistic has $df = 10$. How many participants were used in the entire study?
 - a. 9
 - b. 22
 - c. 11
 - d. 18
11. A researcher reports $t(12) = 2.86$, $p < .05$ for a repeated-measures research study. How many individuals participated in the study?
 - a. $n=11$
 - b. $n=24$
 - c. $n=13$
 - d. $n=25$
12. The distribution of F -ratios is positively skewed.
 - a. True
 - b. False
13. For an experiment comparing more than two treatment conditions you should use analysis of variance rather than separate t tests because of the following.
 - a. Separate t tests would require substantially more computations.
 - b. A test based on variances is more sensitive than a test based on means.
 - c. Conducting several t tests would inflate the risk of a Type I error.
 - d. There is no difference between the two tests, you can use either one.
14. Under what circumstances is the *experimentwise alpha level* a concern?
 - a. whenever you do an analysis of variance
 - b. whenever an experiment compares exactly two treatments
 - c. whenever an experiment involves more than one hypothesis test
 - d. whenever the alpha level is greater than .05
15. An analysis of variances produces $df_{tr} = 3$ and $df_{error} = 24$. For this analysis, what is df_{total} ?
 - a. 26
 - b. 27
 - c. 28
 - d. Cannot be determined without additional information
16. In analysis of variance, the F -ratio is a ratio of the following values.
 - a. sample means
 - b. sample means divided by variances
 - c. two variances
 - d. sample variances divided by sample means
17. In an analysis of variance, which of the following is directly influenced by the size of the sample mean differences?
 - a. SS_{within}
 - b. SS_{total}
 - c. $SS_{between}$
 - d. All three SS values are influenced

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18. In an analysis of variance, which of the following is directly influenced by the size of the sample variances?
 - a. SS_{within}
 - b. SS_{total}
 - c. SS_{between}
 - d. All three SS values are influenced
19. In general, the largest F -ratio will be obtained when the differences between sample means are _____ and the magnitudes of the sample variances are _____.
 - a. Small, small
 - b. Small, large
 - c. Large, small
 - d. Large, large
20. A researcher uses an analysis of variance to test for mean differences among three treatment conditions using a sample of $n = 10$ subjects in each treatment. In this analysis, what value of df would the F -ratio have?
 - a. $df=29$
 - b. $df=2,29$
 - c. $df=2,27$
 - d. $df=3,27$
21. A researcher reports an F -ratio with $df = 2, 36$ for an independent-measures experiment. How many treatment conditions were compared in this experiment?
 - a. 2
 - b. 36
 - c. 3
 - d. 38
22. In analysis of variance, the magnitude of the mean differences from one treatment to another will contribute to the following.
 - a. the denominator of the F -ratio
 - b. both the numerator and the denominator of the F -ratio
 - c. the numerator of the F -ratio
 - d. The sample mean differences do not influence the F -ratio.
23. In analysis of variance, the magnitude of the sample variances will contribute to the following.
 - a. the numerator of the F -ratio
 - b. both the numerator and the denominator of the F -ratio
 - c. the denominator of the F -ratio
 - d. The sample variances do not influence the F -ratio.
24. Under what circumstances are multiple comparison tests necessary?
 - a. when you reject the null hypothesis with exactly two treatment conditions
 - b. when you fail to reject the null hypothesis with exactly two treatment conditions
 - c. when you reject the null hypothesis with more than two treatment conditions

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- d. when you fail to reject the null hypothesis with more than two treatment conditions
- 25. It is impossible to obtain a negative value for an F -ratio.
 - a. True
 - b. False
- 26. Analysis of variance allows researchers to compare several different treatment conditions without conducting several different hypothesis tests.
 - a. True
 - b. False
- 27. In an analysis of variance, SS_{total} will always equal the sum of SS_{between} and SS_{within} .
 - a. True
 - b. False
- 28. In an analysis of variance, df_{between} is determined by counting the number of treatment conditions and subtracting one.
 - a. True
 - b. False
- 29. If $SS_{\text{between treatments}} = 20$ and $MS_{\text{between treatments}} = 10$, then the ANOVA compares two treatment conditions.
 - a. True
 - b. False
- 30. Post tests are not needed if H_0 is rejected in an ANOVA comparing $k = 2$ treatments.
 - a. True
 - b. False
- 31. Post tests are not needed if the decision from an analysis of variance is to fail to reject the null hypothesis.
 - a. True
 - b. False
- 32. Post tests are used to determine how much difference there is between treatment conditions.
 - a. True
 - b. False
- 33. Post test are used to determine which treatment means are significantly different.
 - a. True
 - b. False
- 34. An F -ratio can never be a negative number.
 - a. True
 - b. False
- 35. The results from a two-factor analysis of variance show a significant main effect for factor A and a significant main effect for factor B. Based on this information, you can conclude that
 - a. there must be a significant interaction.
 - b. there probably is a significant interaction.
 - c. the interaction cannot be significant.
 - d. you cannot make any conclusion about the significance of the interaction.

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36. How many separate samples would be needed for a two-factor, independent-measures research study with 2 levels of factor A and 3 levels of factor B ?
- 2
 - 3
 - 5
 - 6
37. For an experiment involving 2 levels of factor A and 4 levels of factor B with a sample of $n = 5$ in each treatment condition, what is the value for df_{within} or df_{error} ?
- 23
 - 29
 - 32
 - 39
38. For an experiment involving 2 levels of factor A and 3 levels of factor B , with a sample of $n = 10$ in each treatment condition, what are the df values for the F -ratio for the $A \times B$ interaction?
- 1, 54
 - 2, 54
 - 3, 54
 - 5, 54
39. The results of a two-factor analysis of variance produce $df = 2, 28$ for the F -ratio for factor A , and $df = 2, 28$ for the F -ratio for the $A \times B$ interaction. Based on this information, how many levels of factor B were compared in the study?
- 1
 - 2
 - 3
 - Cannot be determined without additional information
40. Which of the following is not normally calculated as part of an analysis of variance?
- SS_{total}
 - SS_{E}
 - MS_{total}
 - MS_{E}
41. In an analysis of variance, which of the following is directly influenced by the size of the sample mean differences?
- SS_{within}
 - SS_{total}
 - SS_{between}
 - All three SS values are influenced
42. A two-factor research study has two levels of factor A and two levels of factor B with $n = 10$ participants in each treatment condition. For this study, what is the value for $df_{\text{between treatments}}$?
- 3

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- b. 4
 - c. 9
 - d. 36
43. An analysis of variance comparing three treatment conditions produces $df_{\text{total}} = 24$. For this ANOVA, what is the value of df_{tr} ?
- a. 3
 - b. 21
 - c. 2
 - d. 22
44. Explain what the phrase “95% confident” means when we interpret a 95% CI for μ .
- a. 95% of the observations in the population fall within the bounds of the calculated interval
 - b. In repeated sampling, 95% of constructed intervals would contain the value of μ
 - c. The probability that the mean falls in the calculated interval is 0.95
 - d. 95% of similarly constructed intervals would contain the value of the sampled mean
45. A two-factor ANOVA consists of three separate hypothesis tests.
- a. True
 - b. False
46. The variance within treatments, MS_{within} , is used as the denominator for all three F -ratios in an independent-measures two-factor analysis of variance.
- a. True
 - b. False
47. In a two-factor ANOVA, $MS_{\text{between treatments}} = MS_A + MS_B + MS_{A \times B}$.
- a. True
 - b. False
48. In a two-factor analysis of variance with two levels of factor A and two levels of factor B , what is the relationship among the three F -ratios (A , B , and the $A \times B$ interaction)?
- a. All three F -ratios will have the same values for df .
 - b. The F -ratios for A and B will have the same values for df , but the interaction may have a different df value.
 - c. The F -ratios for A and B will have the same df values but the interaction is guaranteed to have different values for df .
 - d. None of the other 3 choices is correct.

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49. The following table summarizes the results of a two-factor ANOVA evaluating an independent-measures experiment with two levels of factor A, three levels of factor B, and $n = 5$ subjects in each separate sample. Fill in all the missing values in the table. (Hint: Start with the df column)

<u>Source</u>	<u>SS</u>	<u>df</u>	<u>MS</u>	F ratio
Between Treatments	60			
Factor A				$F_A = 5$
Factor B				$F_B =$
A x B	30			$F_{A \times B} =$
Within Treatments			2	
Total				

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Answers:

1. b
2. c
3. c
4. b
5. b
6. a
7. a
8. c
9. c
10. c
11. c
12. a
13. c
14. c
15. b
16. c
17. c
18. a
19. c
20. c
21. c
22. c
23. c
24. c
25. a
26. a
27. a
28. a
29. b
30. a
31. a
32. b
33. a
34. a
35. d
36. d
37. c
38. b
39. b

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- 40. c
- 41. c
- 42. a
- 43. c
- 44. b
- 45. a
- 46. a
- 47. b
- 48. a
- 49. Sum of squares = 10, 20, 48; df = 5, 1, 2, 2, 24, 9; MS = 10, 10, 15; F ratios = 5, 7.5